

# Asifa Firdhouse F

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## Professional Summary

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B.E. Artificial Intelligence and Machine Learning student with hands-on experience in Python, backend APIs, machine learning, and Generative AI. Experienced in building AI-powered applications using FastAPI, RAG, NLP, and data-driven approaches.

## Technical Skills

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- **Programming Language:** Python
- **AI & Machine Learning:** Scikit-learn, NLP, Generative AI, RAG, LLMs, Data Preprocessing, OpenCV
- **Backend Development:** FastAPI, REST APIs
- **Tools & Platforms:** Git, GitHub, VS Code
- **Database:** MySQL

## Education

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**Meenakshi College of Engineering, B.E. – Artificial Intelligence and Machine Learning** Sept 2023 – Present

- **CGPA:** 8.34

## Experience

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**Data Science Intern, THIRANEX Solutions** June 2026 – July 2026

- Performed data collection, preparation, manipulation, and validation while collaborating on data science workflows.

**Full-Stack Developer Intern, IBM** Dec 2025 – Jan 2026

- Developed a Payroll Management System as part of a team-based full-stack development project.

**Python Developer Intern, PS8 Networks** Sept 2024 – Nov 2024

- Developed an AI-based chatbot using Python and NLP techniques.
- Integrated the chatbot with existing platforms using APIs.

## Projects

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### DocuChat – Offline Document Intelligence Platform

- Developed a RAG-based document Q&A system using FastAPI, FAISS, SentenceTransformers, and a local LLM.
- Implemented PDF extraction, chunking, semantic embeddings, vector retrieval, and a React-based document querying interface.

### Smart Predictive Maintenance System for Surgical Robots

- Developed a FastAPI backend for sensor data processing and real-time machine learning predictions.
- Integrated NodeMCU sensors to collect temperature, vibration, and current data for equipment monitoring.
- Implemented an Explainable Boosting Machine (EBM) model for predictive maintenance and ML integration.

### DeepFake Detection

- Contributing to an AI-based system for detecting manipulated media and identifying potential cyber threats.
- Developing backend components for threat detection, phishing analysis, and automated report generation.

### Face Recognition Attendance System

- Developed a real-time face recognition attendance system using Python and OpenCV, with student registration, attendance tracking, CSV export, and an admin interface.

## Certifications

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**Generative AI – GUVI • Full-Stack Development – Infosys • Python – GUVI • Data Science – Infosys • Big Data – Infosys**

## Hackathons

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Smart India Internal Hackathon 2024 • MSME Hackathon 2025 • SAWIT AI Hackathon 2025 – GUVI/HCL